Residual intersection properties for the Mathieu group of degree twenty-two

Human Compression project: bounded application

We compute every residual shadow property of the fifty previously classified locally two-transitive M_{22} geometries of ranks three through five. The complete residue theorem in *Shadows and residues in finite coset geometries*, its sparse right-coset implementation proved in *Shadows of the Hall–Janko geometries*, and the accepted phase-seven complete census are retained in full. This is an application of those common proofs; no subgroup census is repeated.

1 Actual source families and retained hypotheses

The fixed supplementary archive accompanying Jacobs and Leemans, *An algorithmic analysis of the intersection property*, LMS J. Comput. Math. 7 (2004), 284–299, has 35, 22 and 9 active entries in `m22.rk3`, `m22.rk4` and `m22.rk5`. These counts use its actual final reorderings, rather than construction positions. An ambient isomorphism identifies their generated group with the accepted simple group of order 443520. Every entire parabolic family is matched under automorphisms and permutations of types, by successive transporters in the stabilizers of already placed subgroups. The completeness argument for this matching is the one stated in the retained Hall–Janko note. The check uses actual subgroup equality, not subgroup-order profiles.

Write R_i for entry i of the accepted phase-seven witness file. The active families cover exactly $R_8, \dots, R_{32}$ in rank three, $R_{33}, \dots, R_{48}$ in rank four, and $R_{49}, \dots, R_{57}$ in rank five. The following classes each occur twice in the active archive:

rank three	$8, 9, 10, 20, 22, 23, 25, 27, 28, 29$
rank four	$33, 34, 43, 44, 45, 46.$

All other classes occur once. The full 66-entry active/construction/prior correspondence is retained as required data. None of these identifiers is a historical printed geometry label.

The final source lists omit construction entry 4 in rank three, entries 16–19 in rank four and entry 5 in rank five. The earlier exact archive comparison records their failures of local two-transitivity. They are not newly tested in this packet and are not active inputs. In contrast, the sixteen repeated active families above are not rejected geometries; their removal only avoids repeating an isomorphic shadow computation.

The complete earlier census establishes flag transitivity, firmness, residual connectivity, residual weak primitivity, local two-transitivity and $(\mathrm{IP})_2$ under its explicit automorphism/type convention. An isomorphism of the whole subgroup family induces an incidence isomorphism on cosets and every residue, so all these hypotheses transfer to the matched archive entries. The present run independently checks all flag-extension equalities, residual parabolic generation, panel sizes and two-transitivity, and both rank-two shadow directions. Weak primitivity is explicitly retained through the exact census equivalence.

2 Complete residual decisions

Theorem 1. *The counts for the fifty distinct inputs are as follows. Each triple lists IP, weak IP alone, and failure of weak IP, respectively.*

Input rank	*number*	$(\mathrm{IP})_3/(\mathrm{WIP})_3$	$(\mathrm{IP})_4/(\mathrm{WIP})_4$	*full IP/WIP*
3	*25*	$(5, 8, 12)$	—	$(5, 8, 12)$
4	*16*	$(5, 4, 7)$	$(3, 1, 12)$	$(3, 1, 12)$
5	*9*	$(0, 1, 8)$	$(0, 1, 8)$	$(0, 1, 8)$

For rank-five inputs the $(\mathrm{IP})_5/(\mathrm{WIP})_5$ *triple is also* $(0, 1, 8)$*. All fifty geometries satisfy* $(\mathrm{IP})_2$*.*

Proof. For every proper set J of fixed types, take the actual stabilizer H_J and the remaining parabolics $H_J \cap H_i$. The already established flag transitivity identifies every residue of this type with this base residue. For each target type, the common theorem reduces the required object–flag comparisons to its complete double-coset and witness search. We use exactly the sparse Hall–Janko algorithm: nonempty shadows are sets of canonical right-coset representatives, and the empty flag is the whole universe, handled by its proved cardinality test. The mathematical program is unchanged; only the descriptive retained-census field names the M_{22} premise. Trivial panel stabilizers remain allowed.

The run covers 694 base residues including panels. The 510 base residues of rank at least two give 1348 directional truth values. Every negative value includes an actual normalizing permutation and failed intersection after the complete witness search. For each rank k, conjunction over all target types and all base residues gives $(\mathrm{IP})_k$; choosing a successful type separately for each residue gives $(\mathrm{WIP})_k$. The actual number of residues represented by a base flag is $[G : H_J]$. Full IP and weak IP are the respective conjunctions over ranks two through the input rank. Panels are automatic for these properties and rank zero is excluded. Applying these quantifiers to the complete saved decisions gives the table. □

The full positive and weak-only classes, in the retained census numbering, are

rank	full IP	full weak IP alone
3	$10, 23, 28, 31, 32$	$9, 12, 14, 19, 20, 21, 24, 27$
4	$36, 40, 45$	38
5	$\varnothing$	54.

These aggregate values agree with every M_{22} row of source Tables 1, 3, 5 and 7. The identifiers in Tables 2, 4, 6 and 8 come from an earlier classification. Neither matching counts nor the displayed truth profiles establishes that historical correspondence; it remains a separate task.

The published shadow algorithm and source results belong to Jacobs and Leemans. The complete earlier census and its subgroup data, the full six-page residue proof, and the full three-page sparse-adapter note are retained mathematical premises, not free imports. GAP 4.15.1 exact group, coset and automorphism routines remain trusted. There is no new Lean or whole-original recovery claim and no numerical factor. Primary Section 6 contains executable Magma; future code costs must be treated symmetrically.